SCH4U DIY Thesis Lab

A Non-Standard Vinegar Cell and the Nernst Equation

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Purpose

- To compare the measured voltage of non-standard vinegar-based electrochemical cells using a copper cathode against the theoretical cell potentials predicted by the Nernst equation.
- To calculate the pH of the vinegar electrolyte from experimental cell potentials and compare these results to direct pH measurements.

Introduction

Electrochemical cells convert chemical energy from spontaneous redox reactions into electrical energy. In this non-standard cell, hydrogen ions from an acetic acid solution (vinegar) undergo reduction at an inert copper cathode surface, while various metals (zinc, aluminum, or tin) undergo oxidation at the anode. Because the hydronium ion concentration in weak acetic acid is significantly lower than the standard 1.0 M, the resulting cell potential is lower than the standard reduction potential, a relationship modeled quantitatively using the Nernst equation.

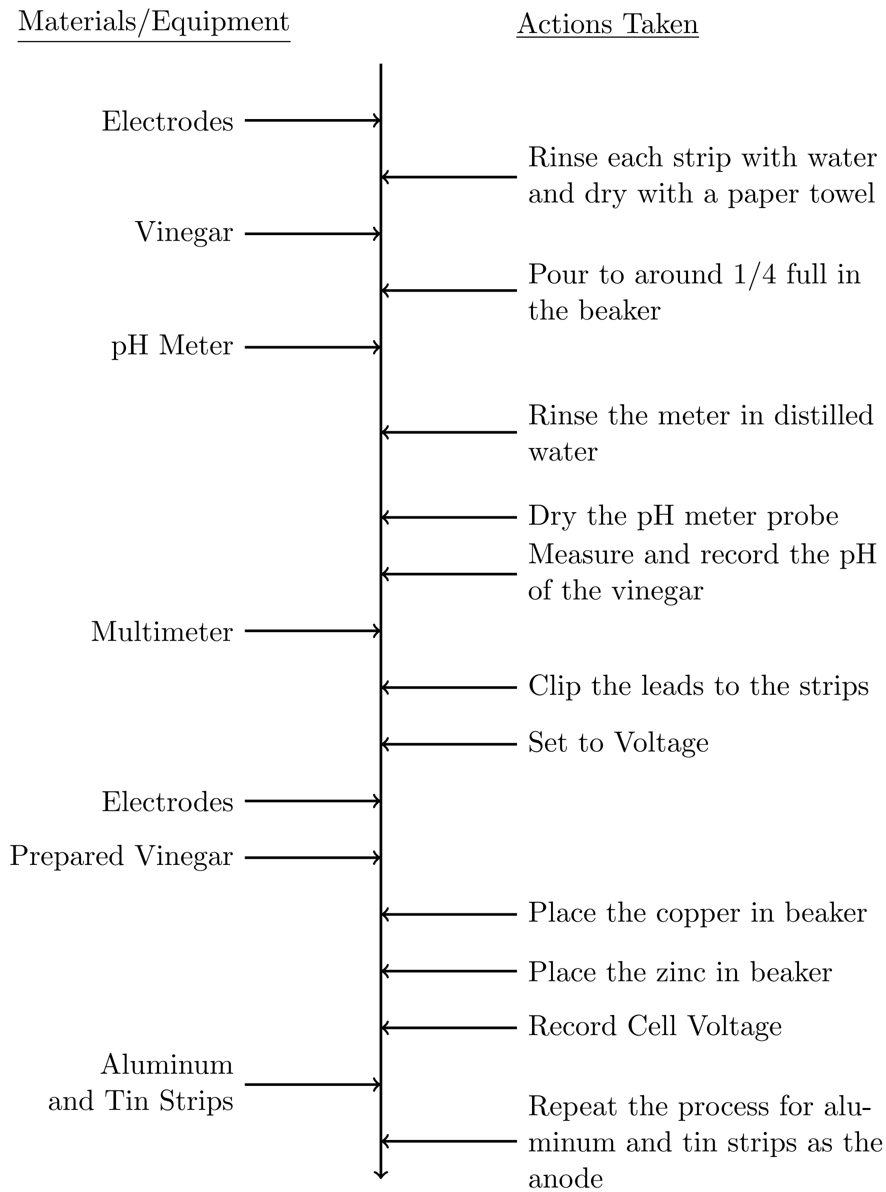
The experimental procedure involved preparing various metal anode strips and pairing each with a common copper cathode in a beaker filled with white vinegar. A multimeter was utilized to measure the resulting cell potentials, and a digital pH meter was used to record the baseline pH of the vinegar electrolyte. These measurements allowed for a direct mathematical comparison between the theoretical and experimental cell potentials.

Safety goggles were used to protect against splashing of vinegar into eyes.

Materials

- 100 mL beaker
- Copper strip (electrode)
- Zinc strip (electrode)
- Aluminum strip (electrode)
- Tin strip (electrode)
- White vinegar (acetic acid solution)
- Multimeter with leads
- Alligator clips
- Distilled water
- Safety goggles

Procedure Flowchart



Observation

Table 1: Voltage and Qualitative Measurements

Cell Combination	Voltage (V)	Observations
Copper / Zinc	0.94	Lost a layer Looked cleaner/shinier
Copper / Aluminum	1.52	
Copper / Tin	0.39	

Table 2: pH of Vinegar

Vinegar Sample	pH
Vinegar	2.4

Analysis

Step 1: Prediction of Cell Voltages

To predict the cell potentials (E_{cell}) under non-standard conditions, the Nernst equation at 25°C was utilized:

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0592\,\text{V}}{n}\log(Q)$$

where Q is the reaction quotient, defined as:

$$Q = \frac{[\text{M}^{n+}]}{[\text{H}^+]^2}$$

Calculation of Hydrogen Ion Concentration

Given the measured pH of 2.4 (Table 2), containing one significant digit in its fractional part, the hydronium concentration was determined as:

$$[\text{H}^+] = 10^{-\text{pH}} = 10^{-2.4} = 4 \times 10^{-3}\ \text{M}$$

Sample Calculation: Copper-Zinc Cell

Using $E^\circ_{\text{cell}} = +0.76$ V, $n = 2$, and assuming an estimated trace metal ion concentration of $[\text{Zn}^{2+}] = 1.0 \times 10^{-5}$ M:

$$\begin{aligned}
E_{\text{cell}} &= E^\circ_{\text{cell}} - \frac{0.0592}{2} \log\left(\frac{[\text{Zn}^{2+}]}{[\text{H}^+]^2}\right) \\
E_{\text{cell}} &= 0.76 - 0.0296 \log\left(\frac{1.0 \times 10^{-5}}{(4 \times 10^{-3})^2}\right) \\
E_{\text{cell}} &= 0.76 - 0.0296 \log(0.625) \\
E_{\text{cell}} &= 0.76 - 0.0296(-0.20) \\
E_{\text{cell}} &= 0.76 + 0.0059 \\
E_{\text{cell}} &\approx +0.77 \text{ V}
\end{aligned}$$

The predicted non-standard cell potential for the copper-zinc cell was determined to be +0.77 V.

Step 2: Prediction of pH from Measured Voltage

The Nernst equation was rearranged to solve for the pH of the electrolyte solution using the measured experimental cell potentials.

Mathematical Derivation

Starting with the non-standard relation:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0592}{2}\log[\text{M}^{2+}] + 0.0592\log[\text{H}^+]$$

Substituting $\text{pH} = -\log[\text{H}^+]$:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - 0.0296\log[\text{M}^{2+}] - 0.0592\ \text{pH}$$

Rearranging to isolate pH yields:

$$\text{pH} = \frac{E^\circ_{\text{cell}} - E_{\text{cell}} - 0.0296\log[\text{M}^{2+}]}{0.0592}$$

Sample Calculation: Copper-Zinc System

Using the experimental value $E_{\text{cell}} = 0.94$ V (Table 1) and assuming a trace metal concentration of $[\text{Zn}^{2+}] = 1.0 \times 10^{-10}$ M:

$$\begin{aligned}\text{pH} &= \frac{0.76 - 0.94 - 0.0296\log(1.0\times10^{-10})}{0.0592}\\ &= \frac{0.76 - 0.94 - 0.0296(-10)}{0.0592}\\ &= \frac{0.76 - 0.94 + 0.296}{0.0592}\\ &= \frac{0.116}{0.0592}\\ &\approx \mathbf{1.96}\end{aligned}$$

The pH of the vinegar electrolyte calculated from the copper-zinc cell potential was determined to be 1.96.

Table 3: Calculated vs. Measured Values

Cell System	Theoretical E°_{cell} (V)	Predicted E_{cell} (V)	Measured E_{cell} (V)	Calculated pH from E_{cell}
Copper / Zinc	+0.76	+0.77	0.94	1.96
Copper / Aluminum	+1.66	+1.68	1.52	8.06
Copper / Tin	+0.14	+0.15	0.39	-8.45

Discussion

1. Two significant sources of systematic experimental error were identified during the trials. First, surface oxidation and impurities on the metal strips, noted in the qualitative observations of Table

1, acted as a physical barrier to electron transfer. This surface resistance restricted active electrochemical contact, thereby altering the cell potential from the theoretical Nernstian predictions. Second, the vinegar electrolyte was not refreshed between consecutive trials. This lack of replacement allowed for the progressive accumulation of metal ions and potential localized neutralization near the electrode surface, altering the pH of the boundary layers and shifting subsequent voltage measurements. Furthermore, because the electrode strips were rinsed only with water and not abrasively cleaned with steel wool or sandpaper between trials, pre-existing oxide layers remained on the surfaces, introducing additional resistance.

2. If the investigation were to be repeated, several procedural modifications would be implemented to optimize the reliability of the dataset. The metal electrodes would be thoroughly sanded with fine steel wool prior to each trial to completely remove passivating oxide layers. Additionally, fresh portions of the vinegar electrolyte would be utilized for each cell pairing, and direct pH measurements would be recorded for every trial to ensure consistency in the baseline hydronium ion concentration.

3. This specific laboratory experiment was chosen because of an interest in the practical construction and optimization of electrochemical cells. The exploration was motivated by a desire to investigate how changes in ion concentration under non-standard conditions shift cell potentials away from standard reduction potentials, and whether the resulting Nernstian deviations could be utilized to mathematically model and resolve unknown solution properties such as pH.

Conclusion

The experimental cell potentials of non-standard vinegar-based cells were determined to be 0.94 V, 1.52 V, and 0.39 V for the copper-zinc, copper-aluminum, and copper-tin systems, respectively, and were successfully utilized to estimate electrolyte pH values. These values were systematically compared against theoretical cell potentials and direct pH measurements, demonstrating deviations that highlight the impact of surface oxidation and concentration shifts in weak acid systems.